SYNCHRONOUS MOTORS UNIT-2

A synchronous motor is quite similar to the alternator in construction. The pole-face winding of the synchronous motor must be more carefully designed since it must perform two functions. It is used in starting the motor and also to prevent hunting.

When balanced polyphase voltages are applied to the stator winding of the synchronous motor, this stator winding will take sufficient current from the supply to build up a flux of sufficient magnitude to generate a back EMF in each winding that is equal and opposite to the applied voltage.

If the supply frequency is f, and if the number of poles is P, the speed of the rotating magnetic field is

$$N_s = \frac{120f}{P} \text{ rpm.}$$ If $P = 2$, $f = 50$ Hz, then $N_s = 3000$ rpm

Principle of working

Synchronous motor works on the Principle of magnetic locking. When the three-phase stator winding is given three-phase supply a rotating magnetic field is produced. In a 2-pole machine, this rotating field is similar to the physical rotation of 2 poles (N_1 and S_1).

When the field winding on the rotor is excited with a dc supply, it also produces 2 poles (N_2 and S_2).

If somehow, the rotor is given an initial rotation at a speed close to N_s in the same direction as the rotating magnetic field, magnetic locking will get established between the stator and rotor poles of opposite polarity. The rotor also rotate at the same speed N_s in the same direction.

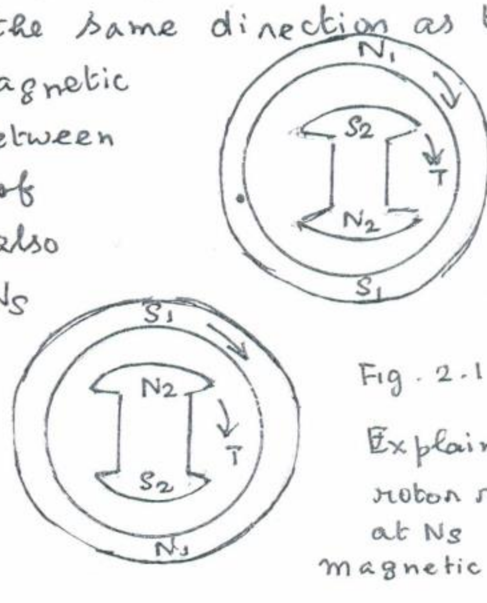


Fig. 2.1

Explaining the rotor rotating at N_s due to magnetic locking.

Why the synchronous motor is not self-starting?

Consider the stator poles N_1 and S_1 and rotor poles
N_2 and S_2. Let N_1 and S_1 be rotating at the
synchronous speed N_S in the clock wise direction.
Let the rotor poles N_2 and S_2 be stationary.

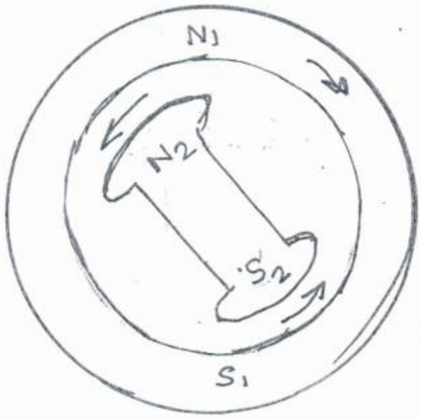


Fig 2.2 a

N_1, S_1: stator rotating field

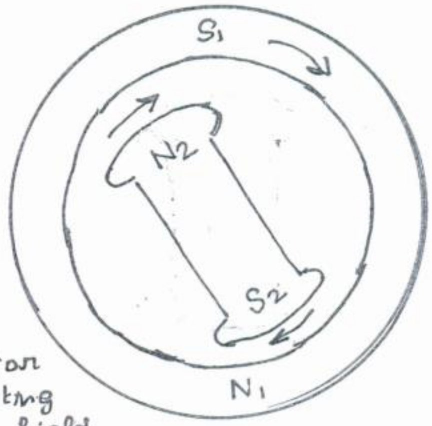


Fig 2.2 b

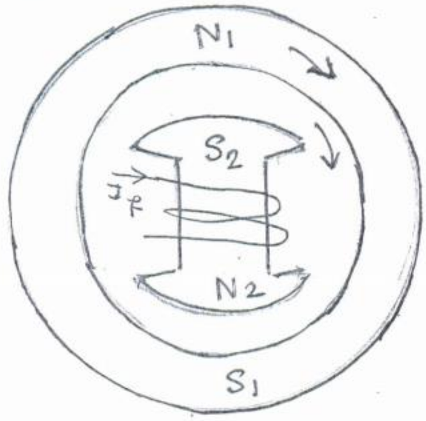
Consider an instant when the stator magnetic field is
vertical as shown in Fig. 2.2a. Let the rotor poles be
arbitrarily positioned as shown. At this instant, like
poles will repel and the rotor will be subjected to an
instantaneous torque in the anticlockwise direction as shown.

Now the stator poles would rotate through half a
revolution in $60/3000 = 20$ m sec. But the rotor, due to
its inertia would have hardly started moving. Then as
shown in Fig 2.2 b, unlike poles attract each other. The
rotor will be subjected to a torque in the clock wise direction.

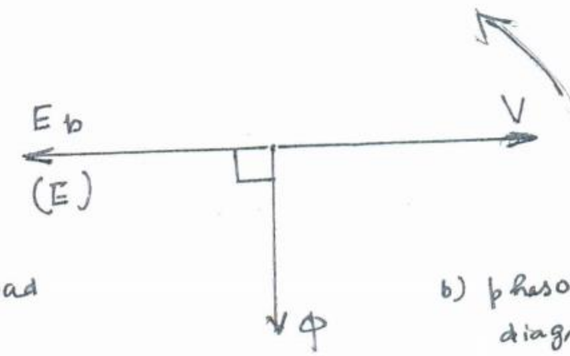
But before this happens, stator poles again change
their positions, reversing the torque exerted on the rotor.
So, the average torque exerted on the rotor is zero, the
rotor remains stationary and hence the synchronous motor
is not self-starting.

In a synchronous motor, damper windings
are placed in the pole shoes (damper bars shorted by
end rings).

On No-load, the magnetic axes of both stator and rotor coincide with each other. The phasor diagram will also be as shown, the field flux leading the back emf by 90°.
Fig. 2.2



a) No-load
$\delta = 0$



b) phasor diagram

Fig. 2.2

When the motor is loaded, the rotor falls back by an angle, δ called load angle, torque angle or power angle. Magnetic locking still exists between the stator and rotor poles and the rotor continues to rotate at synchronous speed, provided δ does not exceed the limit. Under loaded condition, the phasor diagram is as shown in Fig. The resultant torque produced on the rotor depends on δ.

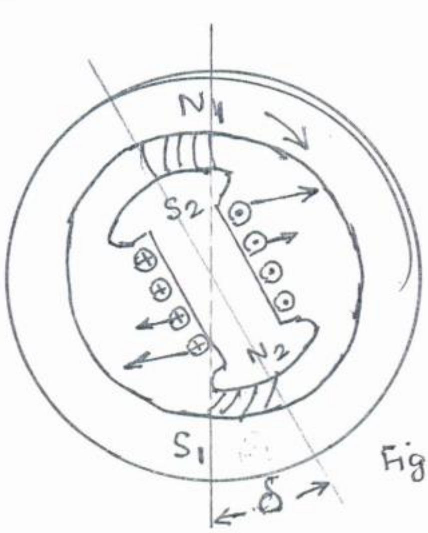


Fig. 2.3

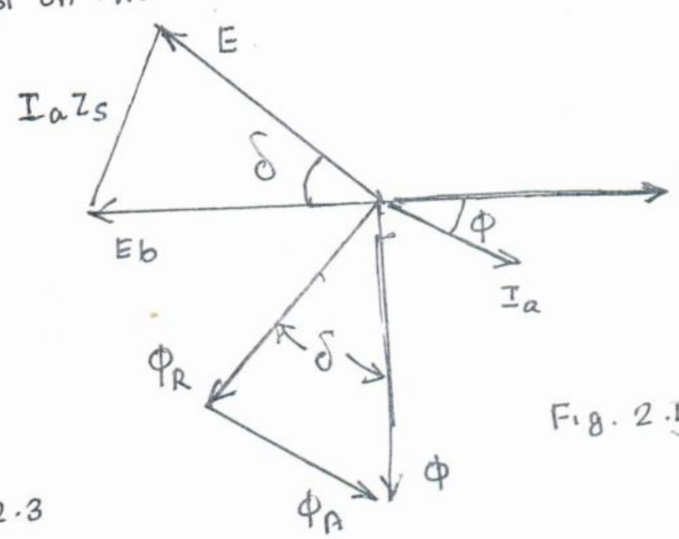


Fig. 2.3b

The phasor diagram can be rearranged as shown in Fig. 2.4

$E = V - I_a Z_s$

$V = V\angle 0$

$E = E\angle -\delta$

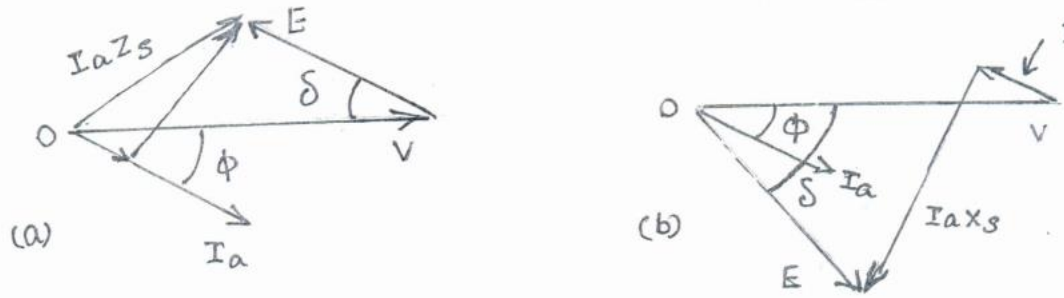


Fig 2.4

<u>The effect of changing the excitation (Load constant)</u>

changes in excitation (changing the Field current) do not affect the torque developed by the motor. The torque is determined by the load.

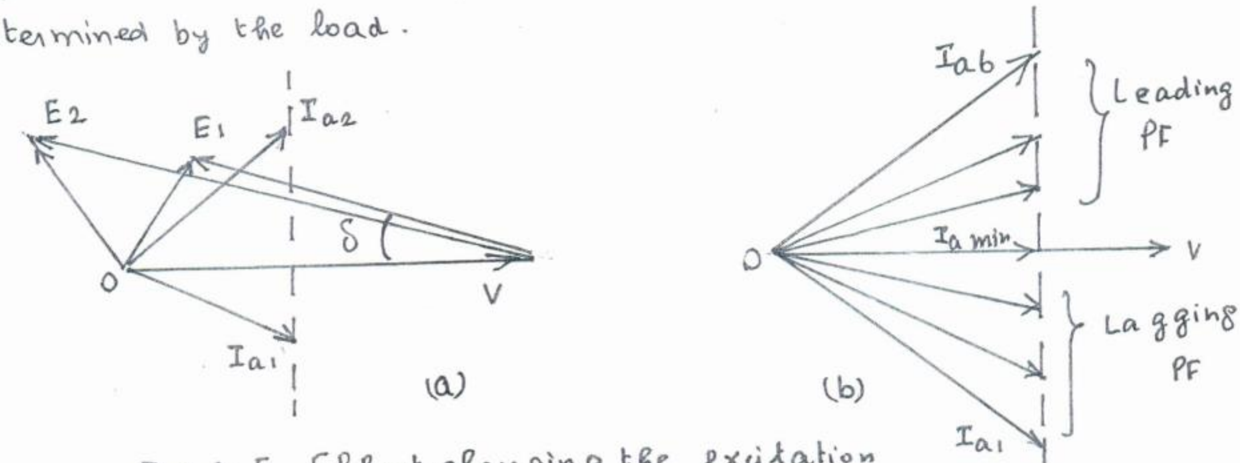


Fig 2.5 Effect changing the excitation

The increase in Field current from I_{f1} to I_{f2}, increases the induced EMF from E_1 to E_2 and the armature current taken by the stator changes from I_{a1} to I_{a2}. The power component of the two currents are essentially equal, but the power factor changes. See Fig 2.5

Normal excitation :	$E = V$	PF lag
Under excitation :	$E < V$	PF lag
over excitation :	$E > V$	PF lead

Minimum armature current occurs at UPF.

The effect of Loading (Excitation constant)

The solid lines in Fig. 2.6 show the phasors for one condition of load on the synchronous motor. An increase in load on the motor will cause an increase in torque angle from δ_1 to δ_2. The dotted lines show the condition of increase in load. change of load changes the armature current from I_{a_1} to I_{a_2}. Note E does not change since excitation is constant.

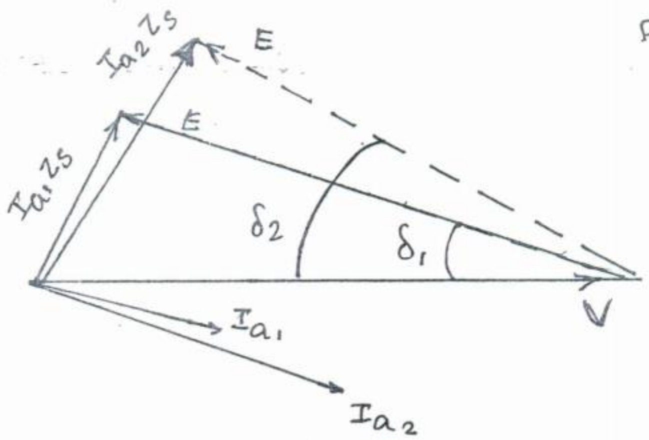


Fig. 2.6 Effect of changing the mechanical load on the motor

V curves and Inverted V curves

From the above discussion about variable excitation of the motor, it is clear that if the field current is increased from a very low (under excitation) to high (over excitation) values; then the armature current I_a initially decreases, becomes minimum at Unity power factor and then again increases.

If the variation of I_a with I_f is plotted, then its shape looks like "V". If such graphs are obtained at various loads, we get a family of V curves.

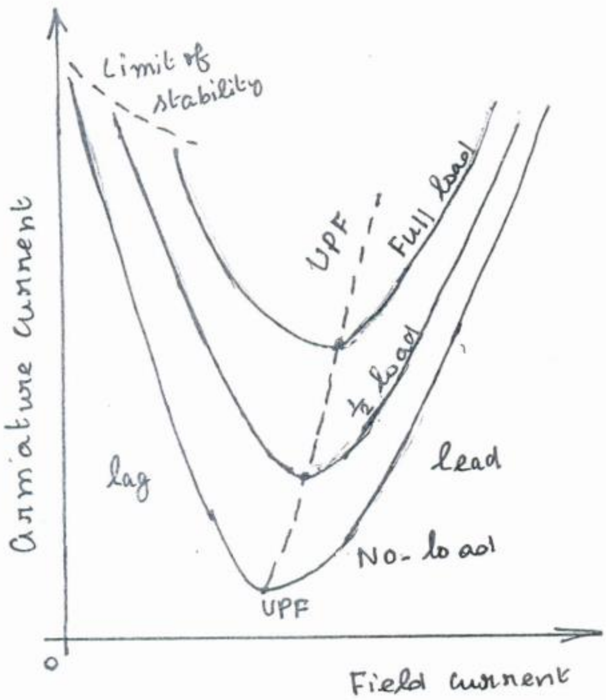


Fig 2.7 V - curves

Note:

i) the shift of the bottom of V, representing UPF at increased loads

ii) No-load armature armature current rises sharply

iii) As excitation is reduced the limit of stability is reached

iv) The lines drawn through points of equal PF at different loads are known as Compounding curves

If the power factor is plotted against field current ($\cos\phi$ Vs I_f), then the shape of the curve looks like inverted "V".

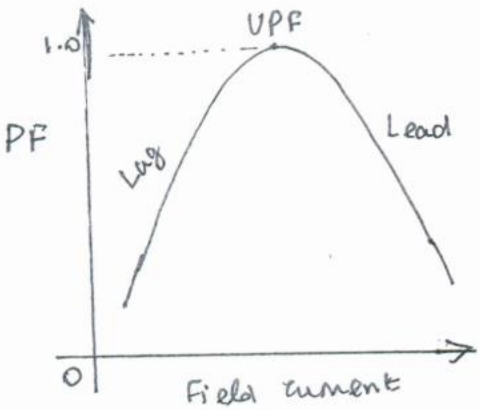


Fig 2.8 Inverted V curves

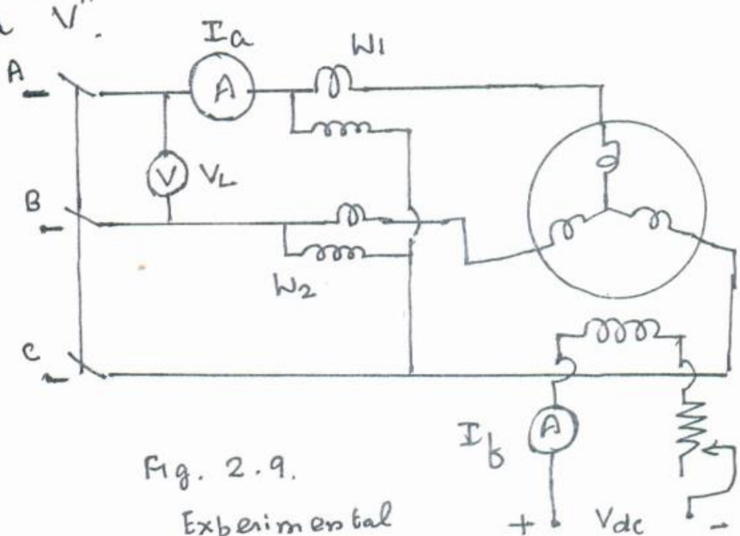


Fig. 2.9. Experimental set up

$$PF = \frac{W_1 + W_2}{\sqrt{3}\, V_L I_a}$$

For a constant load on the motor, if the power input is taken to remain the same, then

$$P_{in} = \sqrt{3}\, V_L I_L \cos\phi = (3 V_p I_p \cos\phi)$$

$I_p = I_a$

So for a constant load, $I_p \cos\phi$ remains constant. Thus, if the PF doubles because of field current adjustment, the armature current, I_a is reduced to half for the power to remain constant.

<u>Power developed by a Synchronous motor</u>

$$I_a = \frac{V - E}{Z_s}$$

$$= \frac{V\angle 0 - E\angle -\delta}{Z_s \angle\theta}$$

$$= \frac{V}{Z_s}\angle -\theta - \frac{E}{Z_s}\angle -\delta - \theta$$

$$= \left[\frac{V}{Z_s}\cos\theta - \frac{E}{Z_s}\cos(\theta+\delta)\right]$$

$$-j\left[\frac{V}{Z_s}\sin\theta - \frac{E}{Z_s}\sin(\theta+\delta)\right]$$

Mechanical power developed $= P_m$

$$P_m = \mathrm{Re}\left[E I_a^*\right]$$

Note: $E = E\cos\delta - jE\sin\delta$

$$= E\cos\delta\left[\frac{V}{Z_s}\cos\theta - \frac{E}{Z_s}\cos(\theta+\delta)\right]$$

$$+ E\sin\delta\left[\frac{V}{Z_s}\sin\theta - \frac{E}{Z_s}\sin(\theta+\delta)\right]$$

$$= \frac{EV}{Z_s}\cos\delta\cos\theta - \frac{E^2}{Z_s}\cos^2\delta\cos\theta + \cancel{\frac{E^2}{Z_s}\cos\delta\sin\theta\sin\delta}$$

$$+ \frac{EV}{Z_s}\sin\delta\sin\theta - \frac{E^2}{Z_s}\sin^2\delta\cos\theta - \cancel{\frac{E^2}{Z_s}\cos\delta\sin\theta\sin\delta}$$

$$= \left[\frac{EV}{Z_s}\cos(\theta-\delta) - \frac{E^2}{Z_s}\cos\theta\right]$$

Then gross Torque $T_g = \frac{P_m}{\omega_s} = \frac{P_m}{(2\pi N_s/60)}$

$$= \frac{9.55\,P_m}{N_s}\ \text{Nm}$$

When R_a is neglected, $\theta = 90^\circ$

$$P_m = \frac{EV}{X_s}\sin\delta$$

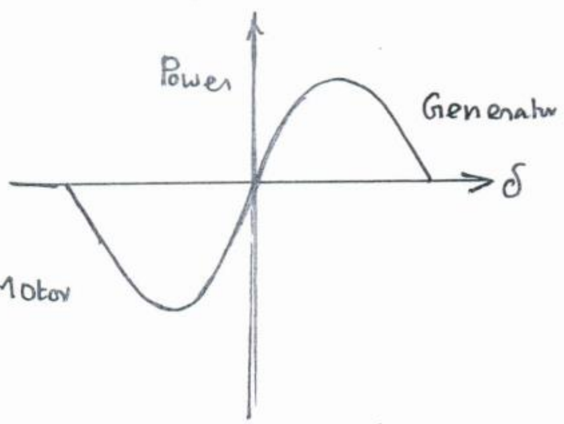
Condition for maximum power developed

The value of torque angle δ for which the mechanical power developed is maximum can be obtained as

$$\frac{dP_m}{d\delta} = \frac{d}{d\delta}\left[\frac{EV}{Z_s}\cos(\theta-\delta) - \frac{E^2}{Z_s}\cos\theta\right]$$

$$= -\frac{EV}{Z_s}\sin(\theta-\delta) = 0 \quad \therefore\ \delta - \theta = 0$$

$$\delta = \theta$$



If R_a is neglected

$$\delta = 90^\circ$$

The corresponding torque is called Pull out torque.

Fig 2.10. Power Angle Curve